

Custom SIEM Dashboard Design & Implementation

Platform: Kibana (ELK Stack)

1. Objective

To design and implement a centralized, interactive SIEM dashboard utilizing Kibana (ELK Stack) that provides real-time situational awareness and visualizes critical security events, specifically focusing on brute-force attacks, privilege escalations, and data exfiltration attempts.

2. Dashboard Components & Visualizations

Widget 1: Brute-Force Activity Heatmap (Geo-Map)

Purpose: To track and visualize the geographical origins of failed authentication attempts.

- **Index Pattern:** `auth-logs-*`
- **Metrics:** Count of events where `event.outcome : "failure"`.
- **Buckets:** Geo-coordinates extracted from source IP addresses using the GeoIP filter.
- **Visual:** A world map highlighting hotspots (red zones) where brute-force dictionary attacks (SSH/RDP) are originating.

Widget 2: Privilege Escalation Monitor (Data Table)

Purpose: To detect unauthorized users gaining administrative or root-level access.

- **Index Pattern:** `winlogbeat-* & auditd-*`
- **Query Logic:** Filters for Windows Event IDs **4672** (Special privileges assigned to new logon) and **4688** (Process creation for sensitive commands like `whoami /priv` or `net localgroup administrators`).

- **Visual:** A real-time data table displaying the *Timestamp, Username, Hostname, and Escalation Command* to allow analysts to investigate suspicious privilege jumps immediately.

Widget 3: Exfiltration Anomaly Tracker (Time Series Area Chart)

Purpose: To identify massive spikes in outbound network traffic that indicate potential data exfiltration.

- **Index Pattern:** packetbeat-* or zeek-*
- **Metrics:** Sum of `network.bytes_out`.
- **Buckets:** Date Histogram (per minute).
- **Query Logic:** Focuses on unusual outbound data volumes over non-standard ports, or massive payloads over DNS (Port 53) and HTTPS (Port 443).
- **Visual:** An area chart with baseline thresholds. Any spike crossing the upper standard deviation boundary triggers a visual red alert on the dashboard.

3. Conclusion

The implemented SIEM dashboard significantly reduces the Mean Time To Detect (MTTD) by transforming raw, unstructured logs into actionable visual intelligence. Analysts can now correlate a brute-force spike with a subsequent privilege escalation event and network exfiltration in a single pane of glass.